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Effects of the 2021 Expanded Child Tax Credit

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Abstract. We analyze the impact of the 2021 Child Tax Credit (CTC) expansion using detailed transaction data from nearly two million individuals and difference-in-difference analyses around monthly CTC payments. We find that recipients significantly increase their consumption in the week following the payment versus the prior week, with more pronounced effects for lower-income recipients and those with more children. We also find a significant reduction in liquidity constraints for lower-income recipients, with reductions in overdrafts, use of high-interest payday loans, and use of gig work for supplemental income. Building on these initial findings, our analysis further delves into more nuanced, policy-targeted aspects: We observe a greater alleviation of liquidity constraints from monthly CTC payments compared with annual tax refunds. In addition, monthly payments decrease consumption volatility and increase stability in consumption levels. These results inform considerations for payment frequency. Furthermore, we provide insights into the income thresholds for the policy's phase-in and phase-out ranges by identifying the most pronounced consumption effects among the lowest-income recipients—who benefit from full refundability under the plan—and noting an absence of significant consumption changes among higher-income recipients. Our results present the first large-scale, transaction-based, empirical archival evidence of the effects stemming from the 2021 CTC amendments and provide insights for policy-related discussions.

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1. Introduction

In this study, we examine the effects of the 2021 expanded Child Tax Credit (CTC) on individual consumption and liquidity constraints. There has been long-standing support for some form of CTC in the United States, as a policy lever used to support families with children. Illustrative of this support, the credit was initiated in 1997, during the presidency of Bill Clinton, and subsequently expanded during the terms of Presidents George W. Bush, Barack Obama, and Donald J. Trump. However, there are open questions about the effects that CTC payments have on families, and ongoing and active debates about how CTC policy should be designed and implemented going forward.

The American Rescue Plan Act (ARPA) of 2021, passed in March, brought several modifications to the CTC. These changes, which we describe below, were in

effect from July 1 to December 31, 2021. Because of the specific changes implemented during this period, the 2021 Act serves as a setting to examine specific questions related to the CTC. Congress actively debated whether to make some of the 2021 changes permanent, with President Joe Biden urging Congress to revive the changes in his 2023 State of the Union address (Adamczyk 2023), and the CTC playing a large role in the 2023 budget negotiations (Fung 2023). The CTC is likely to become a focal point of policy discussions once again, as the expiration of the Tax Cut and Jobs Act (TCJA) looms—the CTC is set to revert to lower levels after the expiration of the TCJA in 2025 (Congressional Research Service 2023).

Three of the most notable ARPA-related changes to the CTC are as follows.¹ First, the legislation altered the payment schedule to provide monthly advance

payments from July through December 2021, rather than disbursing the credit after tax filing in the following year. Second, the legislation introduced a tiered payment system, in which payments were higher for lower-income households. In particular, the legislation increased the amount of the credit from \$2,000 per child per year to \$3,600 for children under age 6 and \$3,000 for children 6 and over, for families with sufficiently low income (e.g., households making less than \$150,000 per year if married filing jointly). Payments were left at the same level as prior to the legislation for families with higher income (e.g., households making roughly \$200,000–\$400,000 per year if married filing jointly). Third, the legislation made the tax credit fully refundable, so that even low- or no-income families would receive the full benefit. Previously, the credit was primarily used to reduce taxes owed, which meant families with low or no income did not receive the full benefit.

We examine the effects of the expanded CTC using data starting from 1.7 billion transactions from bank and credit card accounts for over six million Americans. Our data, described in more detail in Section 2, allow us to identify roughly one million CTC recipients and match them to nonrecipients with similar characteristics.

We provide insight into the effects of monthly CTC payments. An important motivation to change payments to a monthly basis is liquidity provision. The argument is that families will be better able to utilize CTC payments for immediate needs such as food, clothing, and other liquidity needs if payments are made on a monthly basis. We examine the change in recipients' consumption in the week following a CTC payment, relative to the week prior to the payment. Focusing on immediate changes allows us to sharpen the identification of the impact of CTC payments and provide insights into the liquidity provision effect of monthly payments.

We utilize a difference-in-differences (DID) design, comparing postpayment with prepayment behavior for recipients versus nonrecipients, and benchmarking to a prior year (2019, skipping the pandemic year of 2020), to account for any potential disparate calendar-time variation for families with children versus those without. This research design addresses confounding macroeconomic events which affect both CTC recipients and nonrecipients.

We exploit the tiered structure of CTC payments to provide insights into different elements of the program. In particular, we examine the recipients of \$3,000/\$3,600 per year per child (depending on the child's age), whom we call "enhanced recipients," and the recipients of \$2,000 per year per child, whom we call "advance-only recipients." One important design feature of the CTC expansion was the move to monthly payments during the year. Examining advance-only

recipients provides insights into the effects of moving to monthly payments, as payment timing is the only change experienced by this group. Another important feature is the increase in payment amounts, for families with sufficiently low income. Examining enhanced recipients provides insights into the joint effects of monthly payments and increased payment amounts. We also compare the effects across these two groups, to provide insights into the incremental effects of increased payments, and to better understand differences in how recipients in different income ranges respond to monthly payments.

To shed light on whether the intended goals of the 2021 CTC reform—alleviating financial stress and supporting family well-being—are met, we first examine the consumption responses by CTC recipients. We find that both enhanced and advance-only CTC recipients significantly increase total consumption following the monthly payments. Specifically, enhanced recipients increase spending by an average of \$103 in the week after CTC payments relative to the week before, whereas advance-only recipients increase spending by \$22 on average. In addition, we find a stronger spending increase for recipients with more children.

Given policy goals to address families' needs for important categories of consumption, such as food and healthcare, we additionally examine detailed consumption categories. We find that enhanced recipients significantly increase consumption in all categories examined, increasing spending by proportionally more than advance-only recipients in several categories—such as groceries, healthcare, and gas. Both groups similarly increase spending on educational expenses. Together, these results are consistent with monthly payments significantly increasing important categories of consumption, particularly for enhanced recipients.

To illuminate the extent to which CTC payments reduce financial hardship for recipients, we next examine recipients' liquidity following the payments. Consistent with increased consumption, we find a significant reduction in liquidity constraints among CTC recipients. In particular, we find that enhanced recipients experience a significant drop in overdrafts, the use of payday loans, and the use of gig work, typically used for supplemental income (Kousta 2019). For advance-only recipients, we find a reduction in overdraft fees, but no payday loan or gig work effects. We find a significantly stronger drop in liquidity constraints for enhanced than for advance-only recipients for both overdraft fees and payday loan use. Thus, monthly CTC payments appear to provide a significant liquidity benefit, especially to enhanced recipients.

Next, we investigate two additional aspects pertinent to policy considerations. First, we examine how consumption effects vary with respect to recipient income. In particular, we observe the strongest consumption

increases among recipients with estimated incomes below \$45,000 per year, many of whom are likely to benefit from the full refundability of the 2021 CTC expansion. These individuals on average increase weekly spending by \$156 after payment receipt, about \$60 more than other enhanced recipients with incomes above \$45,000. These results highlight the importance of full refundability in ensuring that CTC funds reach the families most in need.

In contrast, we find no increase in consumption for the highest-income advance-only CTC recipients, those estimated to make more than \$150,000 per year. For recipients in the phase-out range of CTC payments, who belong to an even higher income group, we find no increase in consumption. Together, these results suggest that a lower phase-out threshold may be a way to reduce the costs of the CTC program without compromising the goal of facilitating consumption.

Second, we focus more specifically on the effects of payment timing, comparing annual refunds with monthly CTC payments. We argue that monthly payments may help recipients to smooth consumption and avoid binding liquidity constraints. We compare indicators of liquidity constraints in the three months after 2020 tax refunds (paid in early 2021) with those in the three months of monthly CTC payments, controlling for the amount received via either tax refund or CTC payments. We find that, relative to refunds, monthly CTC payments drive a reduction in overdrafts and gig work for both enhanced and advance-only recipients and a reduction in payday loan use for enhanced recipients, incremental to the amount of refund/CTC payment being received. Examining consumption more directly, we find that the volatility of weekly consumption is significantly reduced for monthly CTC payments, relative to a similar period following annual tax refunds.

Finally, we find that both enhanced and advance-only recipients significantly increase payments toward savings, investments, loans, and credit cards, with enhanced recipients experiencing larger increases than advance-only recipients in savings and credit card payments. This suggests that CTC payments go beyond facilitating immediate consumption and alleviating liquidity constraints. Recipients also use these payments to improve their overall financial positions, which may have a longer-term benefit.

Our paper contributes to the nascent literature examining the impacts of the 2021 CTC expansion. This growing body of research delves into a diverse array of CTC-related topics, such as the net cost of the expansion (Goldin et al. 2022), the effects on employment (e.g., Rollet al. 2021, Corinth and Meyer 2022, Ananat et al. 2023), and the effect on child poverty (e.g., Parolin et al. 2021).² Most closely related to our study, Parolin et al. (2023) examine the effect of CTC payments on food hardship and the ability to make rent/mortgage

payments, based on survey data.³ By leveraging detailed transaction data, we identify wider implications of the CTC, showing that it facilitated an easing of liquidity constraints, improvement of financial positions, and increased spending across multiple categories. A distinctive element of our research is how it delves into more nuanced, policy-relevant aspects, specifically examining refundability-relevant and phase-out income levels and the implications of monthly versus once-yearly payments. Thus, our paper stands as a significant addition to the developing CTC literature, deploying granular data to shed light on its real-world effects.

Our study more broadly contributes to the literature examining individual taxpayers. Both economics and accounting researchers are interested in how individual taxpayers respond to varying tax policies. As stated in Yetman and Yetman (2013, p. 1069), “a central issue in tax accounting research is how taxation affects the decisions of businesses and individuals.” Prior research in this area has studied how individual tax compliance is affected by factors such as social norms, enforcement, and tax structure (e.g., Davis et al. 2003, Austin et al. 2020, De Simone et al. 2020). Another theme is the influence of individual taxation on investments (e.g., Falsetta and Tuttle 2011, Falsetta et al. 2013, Stinson et al. 2021). Others have linked personal taxes to corporate policies and financing (Ayers et al. 2003, 2004, 2007; Li et al. 2016; Yost 2018; Armstrong et al. 2019; Hanlon et al. 2021) or examined taxpayers’ consumption responses to tax payments and refunds (e.g., Baugh et al. 2021) and to other government transfers (e.g., Baker et al. 2023b). Building on these analyses, our study leverages the change in the CTC to offer insights into how the timing and amount of tax credits influence taxpayers’ financial behaviors.

2. Data and Summary Statistics

2.1. Individual Transaction Data and Sample

Our data are obtained from a proprietary data vendor, Envestnet | Yodlee, which collects all transactions from bank accounts and credit cards from several partner financial institutions.⁴ The data are deidentified to protect the privacy of individual users, but bank accounts and credit cards belonging to the same user are linked by a unique user identifier.⁵ This allows us to track the transactions by one user across multiple financial institutions and accounts. One limitation is that we cannot link users in the same family if their accounts are not joint—for example, spouses with separate bank accounts and credit cards. In the case of multiple account holders within a single family, we identify the taxpayer/recipient as the account holder who receives the CTC payment. This is likely to add noise to our identification of treatment effects.⁶

For each bank and credit card transaction in the data provided, we have information on the time, amount, direction (i.e., debit or credit), and counterparty. The data also include a description of the transaction, with all sensitive information already masked. We identify CTC recipients as those receiving funds from the Internal Revenue Service (IRS) with transaction descriptions indicating child tax credit.

Panel A of Table 1 describes the sample selection process. Although the Envestnet | Yodlee data include over 10 million individuals, it is crucial that our analyses focus on individuals whose major spending is captured in the data, and whose bank and account records are sufficiently complete, resulting in about 6.5 million individuals. Thus, we start with users for whom the data are more complete and consistent. Our sample period includes the first three CTC payments (July–September 2021) and requires three prior months for matching purposes (April–June 2021).⁷ The database contains 6,471,390 unique users who meet this requirement, with 1,412,065 CTC recipients. We then require the user to have an active data history dating back to April 2019, to allow us to benchmark spending around the CTC to prior year spending. (Note we did not use 2020 because of pandemic-related spending fluctuations.)

For our main analyses, we also exclude recipients with phase-down or phase-out CTC payments or other

CTC amounts that are not typical (e.g., those that are not multiples or combinations of \$166, \$167, \$250, or \$300).⁸ Restricting to typical amounts allows us to estimate the number of children for which recipients receive payments, and whether they are receiving increased payment amounts (enhanced recipients) or advance, but not increased, payments (advance-only recipients). As part of the matching criteria, we require state location data and income tier, an income estimate ranging from 1 to 7 which is estimated by the data vendor based on income and consumption. This results in a sample of 5,433,117 individuals, 1,045,198 of whom are CTC recipients.

To ensure comparability between CTC recipients and nonrecipients, we conduct a parsimonious matching, with replacement. We require the match to reside in the same state as the recipient, and to match on whether they have credit cards covered by the database. We also match based on the following personal characteristics from April through June 2021: income tier, average wage decile, average debt payment decile, and whether the individual has any overdrafts. When multiple nonrecipients meet the criteria, we select the one with the most similar preperiod consumption. We exclude matches where the consumption difference exceeds 50% of the recipient's consumption level (4.2% of matched pairs). In the end, 874,732 (96.3% out of 908,341) enhanced recipients and 120,983 (88.8% out

Table 1. Sample

Panel A: Sample construction						
Sampling criteria	Total individuals	Nonrecipients	All CTC recipients	Enhanced recipients	Advance-only recipients	
Individuals with sufficiently high data quality and transaction records from Apr. through Sept. 2021	6,471,390	5,059,325	1,412,065			
Require account history back to Apr. 2019	5,795,560	4,468,079	1,327,481			
Exclude individuals with phase-down, phase-out, or irregular CTC payments	5,517,494	4,468,079	1,049,415	911,694	137,721	
Require nonmissing state and income information	5,433,117	4,388,519	1,044,598	908,341	136,257	
Match based on individual characteristics	1,991,430	995,715	995,715	874,732	120,983	
Panel B: Distribution of CTC recipients						
Sampling criteria	Enhanced CTC recipients			Advance-only CTC recipients		
	Mean	SD	Median	Mean	SD	Median
\$CTC	479.46	235.71	500.00	278.08	127.55	333.00
#Children	1.80	0.89	2.00	1.65	0.74	2.00
Income tier	4.89	1.70	5.00	6.43	1.10	7.00

Notes. This table describes the main sample for the paper. Panel A describes the sample construction process. We start with all individuals with sufficiently high data quality and transaction records from April through September 2021 (three months before and after the initiation of the 2021 CTC expansion). We then require individuals to have transaction history dating back to April 2019 for benchmarking purposes. To clearly identify enhanced or advance-only CTC recipients, we exclude individuals with phase-down or phase-out CTC payments or irregular CTC payments (extremely large CTC payments or sporadic CTC payments). We further require individuals to have nonmissing demographic data (geographic location and income tier). Lastly, we match CTC recipients to nonrecipients with replacement based on location, data coverage on credit cards, and prior personal characteristics (income tier, overdrafts, wage deciles, debt payment deciles, and average consumption during the three-month preperiod). Panel B provides basic descriptive information for the individuals in the final sample. See Appendix A for detailed variable definitions.

of 136,257) advance-only recipients are successfully matched.

2.2. Data Description and Validation

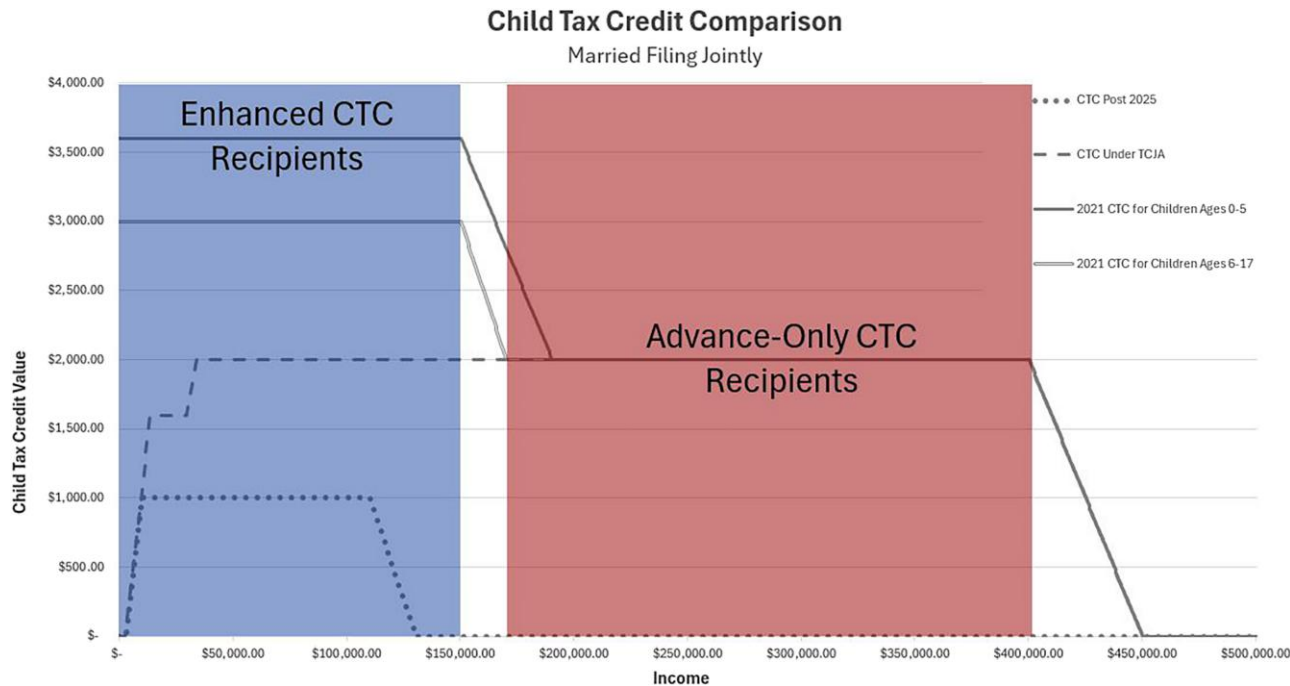
Our final sample of CTC recipients contains roughly one million individuals. According to the IRS, roughly 38 million families received at least one direct monthly CTC payment in 2021 (Internal Revenue Service 2021). Thus, our final matched sample includes roughly 3.3% of CTC recipients nationally. Of the roughly one million CTC recipients in our sample, 87.8% are enhanced recipients, and the remaining 12.2% are advance-only recipients. Using adjusted gross income (AGI) data from the IRS, we estimate that the distribution of enhanced and advance-only recipients in the IRS data is 92.1% enhanced and 7.9% advance-only, comparable to the 87.8%/12.2% distribution in our sample. Because of our data source, and requirements for high-quality data, our sample is tilted toward individuals who utilize banks and credit cards more heavily, which likely explains the slightly higher representation of advance-only recipients in our sample. A limitation of our data is the absence of unbanked individuals, who made up about 5.4% of U.S. households in 2019 according to the Federal Deposit Insurance Corporation (2021). Thus, our results do not speak to the effects of the CTC on the

unbanked and underbanked, and future work that focuses on those recipients will be important to complement the results of our study.

Among the matched CTC recipients, as reported in panel B of Table 1, enhanced recipients receive an average CTC payment of \$479 per month (median = \$500) and advance-only recipients receive \$278 per month (median = \$333). From these monthly payment amounts, we infer the number of children each recipient has and find that the median recipient in each group has two children. By design, enhanced recipients should have lower income than advance-only recipients, as confirmed by the income tier estimated by the data vendor. Specifically, the income tier for the median enhanced recipient is 5, corresponding to the income bracket of \$75K–100K, whereas that for the median advance-only recipient is 7, corresponding to the income bracket of \$150K+. These income tiers align with the expected income tiers, given the phase-out ranges detailed in the policy, as illustrated in Figure 1 for married couples filing jointly.⁹

To evaluate the representativeness of our sample, we examine the geographic distribution of our sample recipients in Figure 2(a), and observe a nationwide distribution, generally representative of the U.S. population (U.S. Census Bureau 2020). Figure 2(b) shows the

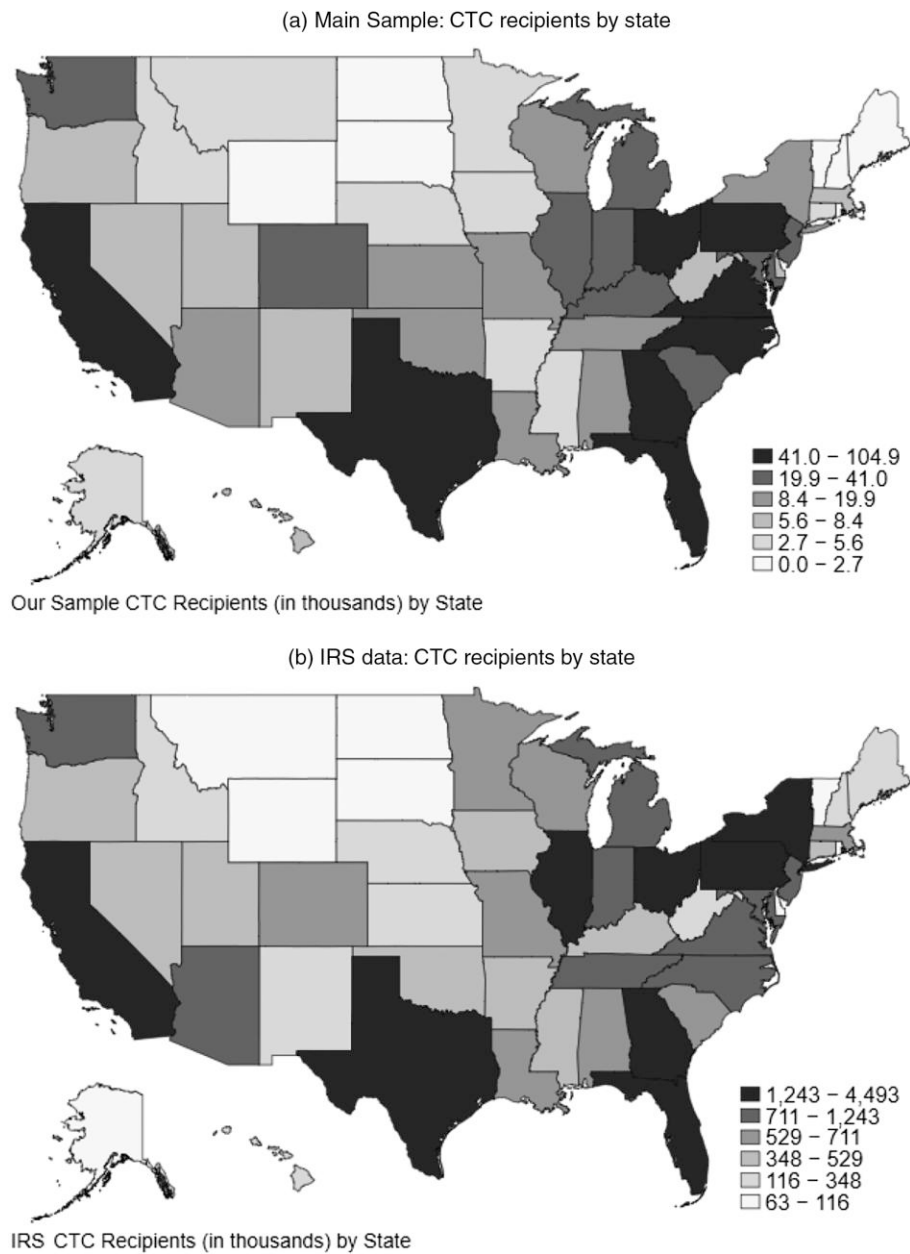
Figure 1. (Color online) Child Tax Credit Expansion



Source. Internal Revenue Code § 24 and ARPA (Public Law 117-2).

Notes. This figure presents CTC payments under the 2021 expansion in a solid line (children under 6) and a double-solid line (children aged 6 or older), for a married couple filing jointly. The blue shaded area on the left illustrates recipients of enhanced CTC in our sample, and the red area on the right is for advance-only recipients. The exact income to payment ranges differ depending on filing status (e.g., single filer, head of household, or married filing jointly). Phase-down begins at \$150,000 for married filing jointly, \$112,500 if filing as head of household, and \$75,000 if single or married filing separately. Phase-out begins at \$400,000 for married filing jointly, or \$200,000 for all other filing statuses. The dashed line represents CTC under TCJA for the tax year of 2025, and the dotted line represents CTC after the CTC provision under TCJA expires at the end of 2025.

Figure 2. Geographic Distribution of CTC Recipients



Notes. This figure presents the geographic distribution of monthly CTC recipients in our sample and compares it to the entire population of CTC recipients. (a) Geographic distribution of CTC recipients in our sample. (b) Distribution of the whole population of recipients according to IRS data.

geographic distribution of the entire population of CTC recipients as reported by the IRS. Although there are some differences—for example, slightly more recipients in our sample are from North or South Carolina, and slightly fewer from the Northeast—the overall distribution is similar. A chi-square test shows no statistical difference between the two distributions.

By design, the CTC is paid out in the middle of the month starting in July 2021, so most payments arrive in bank accounts during the 12th–16th of the month. The exact timing varies across months, largely because of

whether the 15th of the month falls on a weekday or not. There is some slight timing variation across individuals within the same month, as banks process the payments at different paces.

2.3. Outcome Variables and Summary Statistics

To address our research questions about the impact of the CTC expansion on individuals, we examine several variables that proxy for consumption and liquidity constraints. First, to understand how families use CTC payments, we examine total consumption, which is defined

as total weekly spending, excluding transactions that are made in the following categories as classified by the data vendor: loans, mortgages, credit card payments, savings, retirement contributions, securities trades, taxes, and transfers. Weeks are defined relative to CTC payment dates, focusing on one week before, days $[-7, -1]$, and one week after, days $[0, 6]$, relative to each CTC payment, from July through September 2021. To gauge the comparability between CTC recipients and their matched nonrecipients, we report the summary statistics for $[-7, -1]$ in Table 2 and find the average spending is slightly higher for enhanced recipients and lower for advance-only recipients than their respective matched nonrecipients. The difference is statistically significant, but the economic magnitude is small (-0.27% and 0.67% , respectively).

To provide more granular evidence on consumption patterns, we also examine spending in several specific categories as defined by the data vendor: groceries, other food/dining (including fast-food establishments, restaurants, etc.), education, healthcare, gas, and cash (ATM withdrawals to proxy for cash spending). Appendix B lists the most frequent merchants included in each consumption category. These merchants appear in the expected categories, which suggests the category-specific consumption largely captures its intended spending.¹⁰ For the full sample, including matches, the categories with the highest average spending and the most frequent spending are groceries and other food/dining, consistent with frequent spending on food items. The category with the lowest, and least frequent, spending among those examined is education.

To examine the impact of the CTC on families' liquidity, we construct three proxies for liquidity constraints. The first proxy is the occurrence of overdraft or nonsufficient funds charges on bank accounts (*Overdraft(D)*: a dummy indicator for the presence of such charges, which is defined in detail in Appendix A). These charges indicate insufficient funds to cover transactions. On average, overdrafts are infrequent. Only 1% of enhanced recipients and their matches incur overdrafts, and the percentage is even smaller for advance-only recipients and their matches.

The second proxy for liquidity constraints is the incidence of payday borrowing (*Payday_Loan(D)*: a dummy indicator for the presence of payday borrowing), which is defined in detail in Appendix A.¹¹ Because payday loans charge high interest or fees, they are usually one of the last resorts for individuals in liquidity shock to bridge financial shortfalls (e.g., Morse 2011). A typical two-week payday loan equates to an annual percentage rate of almost 400% (Consumer Financial Protection Bureau 2024). Similar to overdrafts, payday borrowing is rare on average. In a typical week centered around CTC payments, less than 1% of enhanced recipients and their matches resort to payday loans, and the percentage is even smaller for advance-only recipients and their matches.

The third proxy for liquidity constraints is gig work participation (*Gig_Work(D)*: a dummy indicator for the presence of gig work participation)¹², as identified by the income from 48 gig platforms such as Uber and Upwork.¹³ Research has shown that workers are more likely to take gig work after facing income and asset

Table 2. Summary Statistics

Variables	Enhanced subsample						Advance-only subsample					
	CTC recipients		Matches		\$	%	CTC recipients		Matches		\$	%
	Mean	S.D.	Mean	S.D.			Mean	S.D.	Mean	S.D.		
Tot_Con\$	1,272.00	1,347.19	1,268.51	1,428.05	−3.485***	−0.27%	1,906.31	1,941.34	1,919.01	1,979.36	12.701***	0.67%
Groceries\$	105.13	138.09	92.14	127.39	−12.998***	−12.36%	129.27	170.00	122.58	160.88	−6.697***	−5.18%
Other_Food_Dining\$	109.17	136.17	98.15	134.43	−11.023***	−10.10%	130.03	173.12	127.52	170.82	−2.511***	−1.93%
Education\$	2.87	16.94	1.62	12.62	−1.252***	−43.66%	5.46	24.31	2.72	16.91	−2.737***	−50.17%
Healthcare\$	17.65	50.96	17.63	51.94	−0.020	−0.11%	24.11	63.13	24.15	63.20	0.037	0.15%
Gas\$	55.93	108.92	49.84	107.43	−6.090***	−10.89%	58.71	122.71	60.00	126.12	1.284***	2.19%
Cash\$	37.58	117.80	41.65	127.19	4.067***	10.82%	35.20	117.90	45.90	138.20	10.696***	30.38%
Overdraft(D)	0.01	0.11	0.01	0.10	−0.002***	−15.61%	0.00	0.06	0.01	0.07	0.001***	24.17%
Payday_Loan(D)	0.01	0.09	0.01	0.08	−0.001***	−18.72%	0.00	0.04	0.00	0.05	0.001***	67.70%
Gig_Work(D)	0.02	0.13	0.02	0.13	−0.000***	−2.39%	0.01	0.10	0.01	0.12	0.004***	32.73%
Saving\$	2.13	16.09	1.94	15.61	−0.197***	−9.24%	4.74	24.90	2.77	19.09	−1.971***	−41.55%
Invest\$	8.55	60.55	10.65	69.33	2.100***	24.57%	25.76	109.82	23.26	105.48	−2.500***	−9.71%
Loan\$	170.04	476.18	176.82	494.48	6.772***	3.98%	326.70	757.71	339.40	753.33	12.692***	3.88%
CC_Payment\$	415.83	1,101.23	471.68	1,172.34	55.849***	13.43%	1,169.21	2,029.42	1,061.99	1,867.40	−107.225***	−9.17%

Notes. This table reports the mean and standard deviation of selected key variables for enhanced recipients, advance-only recipients, and their matched nonrecipients during the week before the receipt of CTC payments. We then report the differences in dollar amount (statistical significance is tested using two-tailed *t*-tests) and percentage. We winsorize all variables except dummy, count, or logged variables at 1% and 99%. Please refer to Appendix A for detailed variable definitions.

***, **, and * stand for statistical significance at the 1%, 5%, and 10% level, respectively.

declines, suggesting that the need to earn extra income from gig jobs reflects liquidity constraints (Koustas 2019). On average, enhanced recipients and their matches are slightly more active participants in the gig economy (about 1.7% participation in a given week) than advance-only recipients (about 1.1% participation in a given week). These figures are in line with estimates of gig work participation from other data sources, including IRS data (Garin et al. 2023).

Lastly, Table 2 presents summary statistics for additional financial variables examined in additional analysis: money transferred into savings accounts (*Saving*) or brokerage accounts (*Invest*) and paying down debt (*Loan* for “loans” and “mortgages” and *CC_Payment* for credit card debt). Appendix A provides detailed variable definitions.

3. Empirical Methodology

In order to empirically examine the effects of the CTC on recipients’ consumption and liquidity constraints, we utilize a DID specification. This allows us to capture the differential effect on recipients relative to matched individuals, where the matches are determined as described in Section 2.1. We utilize fixed effects to control for additional factors which might affect consumption and liquidity.

We utilize two samples for our analyses, examining enhanced and advance-only recipients, and their respective matches, separately. The enhanced sample includes one observation for the week before and one for the week after the CTC payment (days $[-7, -1]$ and $[0, 6]$ relative to the payment date) for each enhanced recipient and their match. The advance-only sample is defined similarly. We focus on the two-week window to more clearly identify the direct effects of CTC payments.

The regression models are specified as follows, with Model (1) for enhanced CTC recipients and Model (2) for advance-only recipients,

$$\begin{aligned} Outcome_{i,s,t} = & Week[+1]_{i,t} * Enhanced_CTC_i + Week[+1]_{i,t} \\ & + User_FE_i + State_Month_{s,t} + \epsilon_{i,s,t}, \end{aligned} \quad (1)$$

$$\begin{aligned} Outcome_{i,s,t} = & Week[+1]_{i,t} * Advance - Only_CTC_i \\ & + Week[+1]_{i,t} + User_FE_i + State_Month_{s,t} \\ & + \epsilon_{i,s,t}, \end{aligned} \quad (2)$$

where $Enhanced_CTC_i$ ($Advance - Only_CTC_i$) is an indicator which takes the value of one if the individual receives \$250 or \$300 per child per month (\$166 or \$167 per child per month). The indicator $Week[+1]$ takes the value of one for the weeks after receiving CTC payments. The variables of interest are the interaction terms $Week[+1] \times Enhanced_CTC$ and $Week[+1] \times Advance - Only_CTC$. The coefficients on these interaction

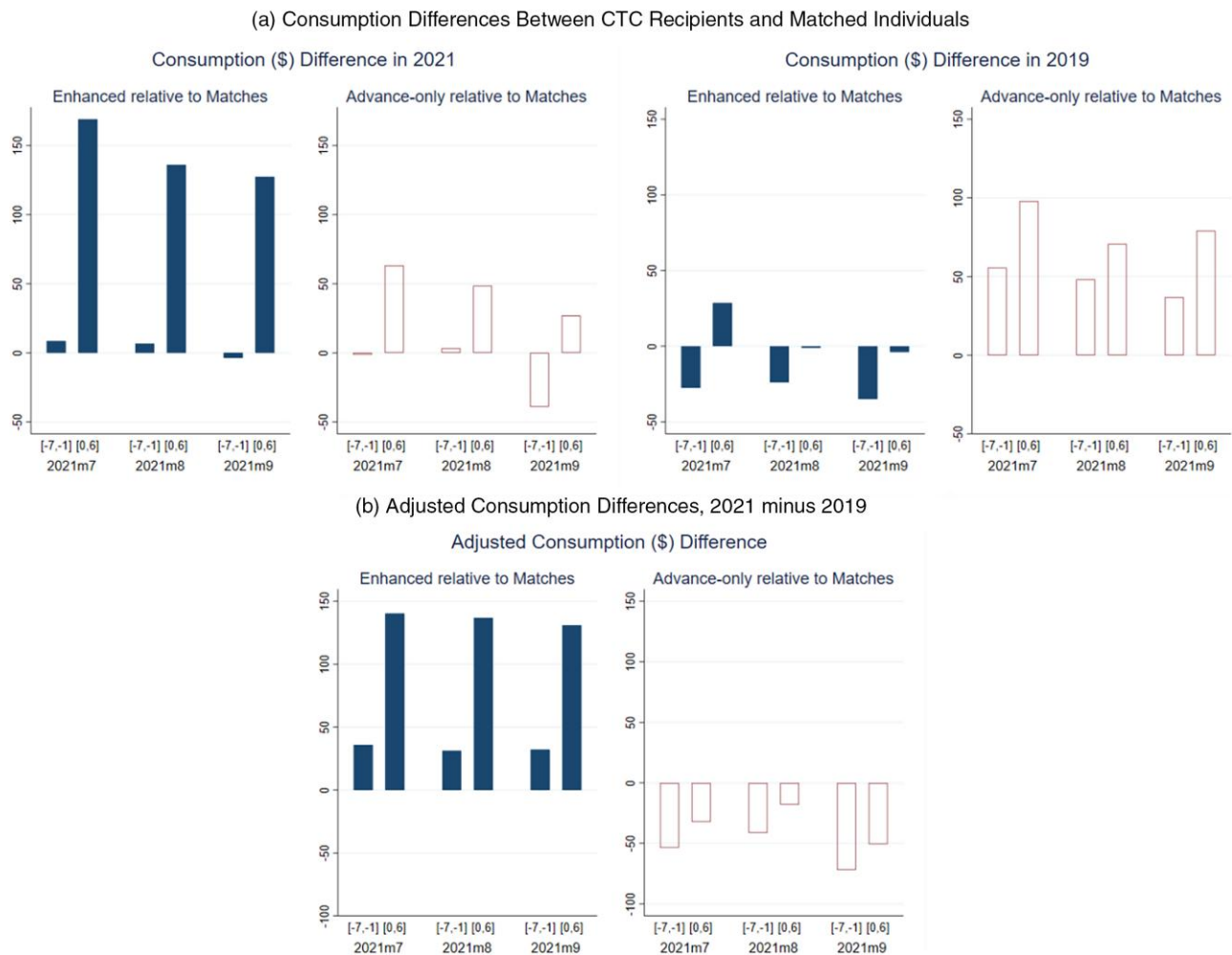
terms are the DID estimate of the CTC impact for enhanced and advance-only recipients, respectively. In particular, they capture how much the outcome variable of interest changes for CTC recipients after receiving the CTC payment, relative to matched individuals. In both models, we include state-month fixed effects to control for state-specific economic conditions. We also include individual fixed effects to control for individual-specific behavior which does not vary with respect to CTC payments. Individual fixed effects absorb standalone CTC recipient indicators. To compare effects for enhanced and advance-only recipients, we compare the DID estimates from Models (1) and (2) by performing stacked regressions that estimate effects for both groups jointly.

Our short-window DID test design offers two major advantages. First, it helps to sharpen the identification of the CTC effect as the narrow window reduces the likelihood that other concurrent events or trends are driving results. Second, the DID design controls for common trends. For macroeconomic events (e.g., inflation) or common seasonality (e.g., some spending is scheduled in the middle of the month) that affects both recipients and matched nonrecipients equally, the effects are differenced out.

However, it is possible that people with children have a different spending timing pattern than individuals without children. If, for example, parents schedule childcare payments in the middle of the month, their total consumption will go up around the 15th, regardless of CTC payments, even relative to individuals who are matched on other economic characteristics. To investigate whether there are systematic calendar-time differences in CTC recipient and match behavior, we plot the differences in weekly consumption between recipients and their matches around CTC payment dates in 2021 and the same dates in 2019 (when no CTC payments were made). These are reported in Figure 3(a). We observe that the postweeks have more spending than the preweeks in both 2021 and 2019, although the increase appears much larger in 2021. To control for the influence of typical calendar-time differences in spending patterns between individuals with and without children, we adjust each outcome by deducting the corresponding value from the same week in 2019. Figure 3(b) plots this adjusted measure for the weeks before and after CTC payments. In this sense, our specification can be considered a triple-difference design, exploiting the differences between pre- and postpayment, recipient and matched nonrecipient, and 2021 and 2019.¹⁴

Because the dependent variables are measured on a weekly basis, consumption, especially category-specific consumption, is highly skewed with frequent zeros. For example, median spending on education is \$0, as educational spending occurs in fewer than 50% of

Figure 3. (Color online) Consumption Benchmarking, 2021 and 2019



Notes. This figure plots the weekly total consumption for enhanced and advance-only recipients, relative to their matches from July 2021 to September 2021. For 2021, $[-7, -1]$ is the week before the CTC payment date and $[0, 6]$ is the week starting from the CTC payment date. For 2019, we use the same dates but shifted two years earlier. (a) Consumption differences between recipients and their matches for 2021 and 2019 separately. (b) Adjusted consumption differences between recipients and their matches, with adjusted consumption defined as 2021 consumption minus 2019 consumption for a given individual.

weeks. To deal with skewness, we take the natural logarithm of one plus the consumption measure in question. We then benchmark against 2019. Thus, the final outcome variables take the form $\text{LnOutcome_Adj}_{i,t} = \text{Ln}(1 + \text{RawOutcome}_{i,t(2021)}) - \text{Ln}(1 + \text{RawOutcome}_{i,t(2019)})$ and $\text{Outcome_Adj}_{i,t} = \text{RawOutcome}_{i,t(2021)} - \text{RawOutcome}_{i,t(2019)}$.¹⁵

Because our primary research design is a DID design, we also examine whether there are parallel trends between CTC recipients and matched individuals in their pre-CTC-payment-week consumption. As seen in Figure 4, preweek consumption for each of the three CTC payment months in our data set moves in parallel for recipients and matches. There are no discernible differential trends across the months, suggesting that the difference-in-differences we observe are driven by CTC payments rather than by differential consumption trends between recipients and their matches.

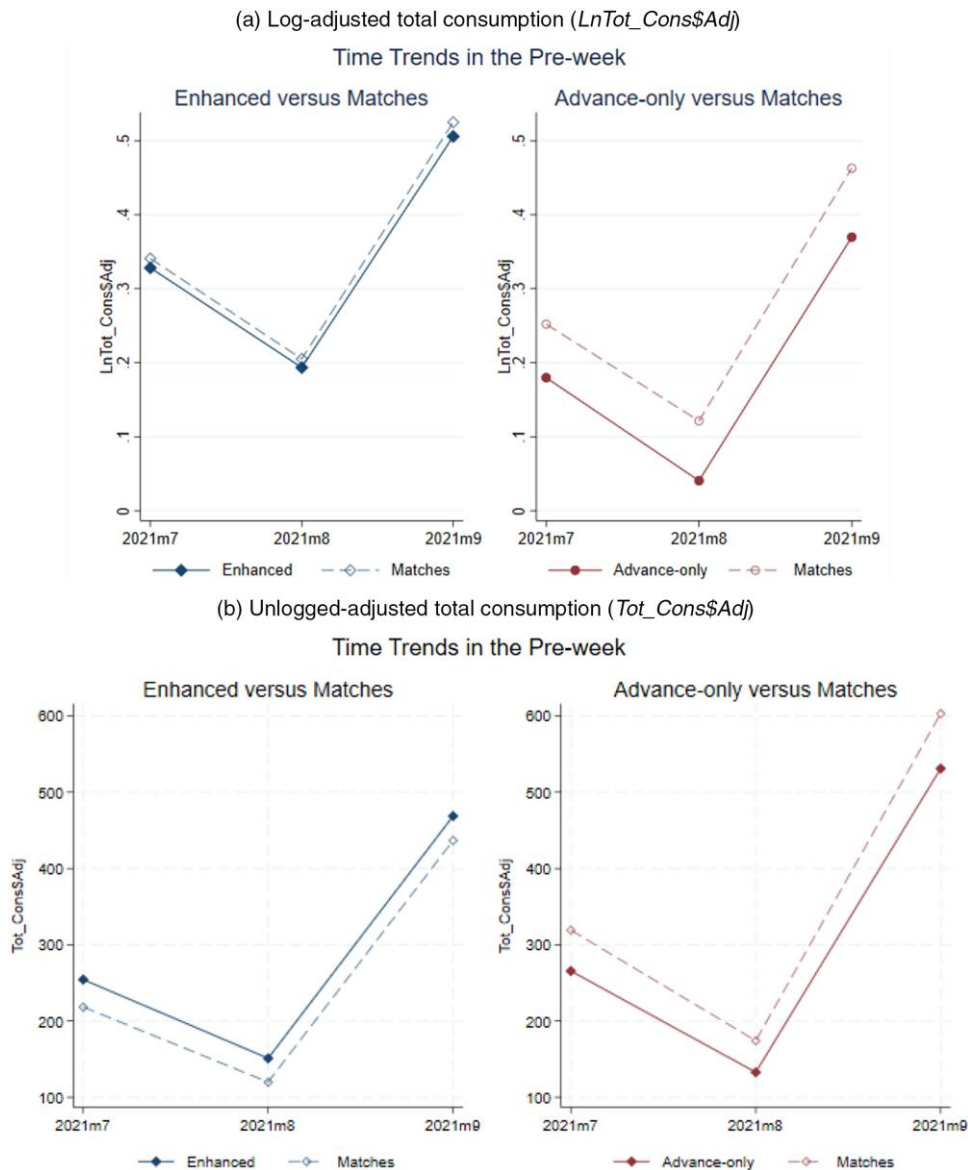
4. Results

4.1. Consumption

Given the goals of the 2021 CTC reform to support families, we first examine the consumption pattern following monthly CTC payments. We examine total consumption as well as consumption in specific categories that align with legislative goals, such as groceries, education, and healthcare. Generally, we expect CTC recipients to increase consumption shortly after receiving CTC payments, with enhanced recipients experiencing a larger increase because they are more likely to be financially constrained and they receive a larger CTC payment.

Table 3 presents results for DID analyses of the effect of CTC payments on recipients' total weekly consumption. Each CTC recipient and matched individual appear six times in the sample: for each of the three months of CTC payments, there is one observation for

Figure 4. (Color online) Preweek Consumption Trends



Notes. This figure plots the preweek consumption for enhanced and advance-only recipients, relative to their matches, in each of the three CTC payment months in our sample. Specifically, we plot the average consumption during $[-7, -1]$ relative to the CTC payment dates (day 0) from July through September 2021 for enhanced recipients (represented by solid diamonds) and their respective matches (represented by hollow diamonds) as well as advance-only recipients (represented by solid dots) and their respective matches (represented by hollow dots). Panel (a) presents the log of total consumption whereas panel (b) presents total consumption in dollar terms, both benchmarked against 2019 (i.e., shifting the measurement window by two years).

the individual's spending in the week prior to the payment, and one observation for the week following the payment. Columns (1) and (2) present results for the log of total consumption, whereas columns (3) and (4) present results for total consumption in dollar terms, both benchmarked against 2019. Results show a significant increase in consumption for both enhanced CTC recipients (columns (1) and (3)) and advance-only recipients (columns (2) and (4)).

In terms of magnitude, the effect is economically larger for enhanced recipients, which is consistent with

these relatively lower income recipients being more liquidity constrained in their spending. Enhanced recipients increase spending by 16% in the week after CTC payments relative to the week before, compared with matched individuals. In dollar magnitude, the average increase is \$103 per week. The average monthly CTC payment for enhanced recipients is \$479 (Table 1, panel B). Thus, we find that the average enhanced recipient increases consumption by 22% of their monthly payment in the week following payment, versus the week prior. These results suggest that the increase in

Table 3. Short-Term Consumption Response to CTC Expansion

Variables	(1) LnTot_Cons\$Adj	(2)	(3) Tot_Cons\$Adj	(4)
Panel A: Baseline results				
Week[+1]	−0.056*** (−30.569)	−0.057*** (−12.033)	−60.294*** (−36.739)	−85.959*** (−15.418)
Week[+1] × Enhanced_CTC	0.161*** (70.226)		102.916*** (51.169)	
Week[+1] × Advance-Only_CTC		0.024*** (3.756)		21.963*** (2.996)
Difference in the DID estimates for enhanced and advance-only		0.137*** (21.097)		80.953*** (10.918)
State-month FE	Y	Y	Y	Y
Individual FE	Y	Y	Y	Y
Observations	10,496,782	1,451,796	10,496,784	1,451,796
Adj R ²	0.411	0.325	0.182	0.149
Panel B: After entropy balancing				
Week[+1]	−0.055*** (−31.409)	−0.054*** (−11.434)	−56.400*** (−34.206)	−83.190*** (−14.578)
Week[+1] × Enhanced_CTC	0.160*** (71.800)		99.022*** (49.080)	
Week[+1] × Advance-Only_CTC		0.021*** (3.297)		19.194*** (2.583)
Difference in the DID estimates for enhanced and advance-only		0.139*** (21.417)		79.829*** (10.618)
State-month FE	Y	Y	Y	Y
Individual FE	Y	Y	Y	Y
Observations	10,496,782	1,451,796	10,496,784	1,451,796
Adj R ²	0.416	0.325	0.183	0.149
Panel C: Number of children				
Week[+1]	−0.056*** (−30.569)	−0.057*** (−12.033)	−60.294*** (−36.739)	−85.959*** (−15.418)
Week[+1] × Enhanced_CTC[1 or 2 Children] (a)	0.149*** (62.445)		87.104*** (42.009)	
Week[+1] × Enhanced_CTC[>2 Children] (b)	0.213*** (57.374)		171.748*** (51.864)	
Week[+1] × Advance-Only_CTC[1 or 2 Children] (c)		0.009 (1.428)		11.588 (1.529)
Week[+1] × Advance-Only_CTC[>2 Children] (d)		0.141*** (12.201)		107.608*** (8.049)
Difference in (a) and (b)	−0.065*** (−18.02)		−84.645*** (−26.93)	
Difference in (c) and (d)		−0.132*** (−11.50)		−96.021*** (−7.28)
State-month FE	Y	Y	Y	Y
Individual FE	Y	Y	Y	Y
Observations	10,496,782	1,451,796	10,496,782	1,451,796
Adj R ²	0.411	0.325	0.182	0.149

Notes. This table reports the impact of CTC expansion on weekly total consumption. Columns (1) and (3) use a matched enhanced CTC sample (874,732 enhanced CTC recipients and their matches) and columns (2) and (4) use a matched advance-only CTC sample (120,983 advance-only recipients and their matches). The sample includes observations from July through September 2021 with one week before the receipt of a monthly CTC payment and one week after. Panel A presents the baseline results, and Panel B presents the results after entropy balancing matching. The dependent variables in columns (1) and (2) ((3) and (4)) are logged (unlogged) weekly total consumption adjusted by its value in the same week in 2019. *Week[+1]* indicates the week starting from the day with the CTC payments. *Enhanced_CTC* and *Advance-Only_CTC* indicate enhanced CTC recipients and advance-only recipients, respectively. The row “Difference in the DID estimates for enhanced and advance-only” reports the results of tests for the coefficient difference between the enhanced CTC and advance-only CTC samples using two-tailed *t*-tests, based upon stacked regressions which estimate effects for both groups jointly. Panel C presents the cross-sectional tests by the number of children for enhanced and advance-only CTC recipients, respectively. Please refer to Appendix A for detailed variable definitions. We control for individual fixed effects (FE) and state-month fixed effects. *T*-statistics based on standard errors clustered by individual are reported in parentheses.

***, **, and * stand for statistical significance at the 1%, 5%, and 10% level, respectively.

payment amount and the change in timing together facilitate immediate consumption on the part of recipient families. In comparison, advance-only recipients increase weekly consumption by 2.5% and \$22 on average. Compared with an average payment of \$278, this estimate suggests that the average advance-only recipient consumes roughly 8% of their monthly payment in the week following payment relative to the week prior.¹⁶ Thus, monthly payments, without any change to payment amount, facilitate immediate spending even for higher-income households.

Given the observable differences between recipients and nonrecipients as shown in Table 2, we perform entropy balancing matching to ensure covariate balance between treatment and control individuals. Specifically, we balance for consumption in total and by category, the three proxies for liquidity constraints, and the four variables of financial positions in the week prior to the first CTC payments. We repeat the analyses after balancing and report the results in panel B of Table 3. The results are similar to those obtained without matching. Enhanced and advance-only recipients increase spending by 16% and 2.0% (\$99 and \$19 in dollar terms), respectively, in the week after CTC payments relative to the week before, compared with matched individuals. The similarity of estimates with and without entropy balancing indicates that our results are not confounded by the observable differences between CTC recipients and nonrecipients.¹⁷

Next, we examine the consumption effects of the 2021 CTC expansion separately for families with one to two children and families with more than two children. We expect a stronger effect for families with a larger number of children, given that the size of the payments is based on the number of children. This serves as a validation that our results are driven by CTC payments. We modify Equations (1) and (2) to separately examine recipients with one or two children ([1 or 2 Children]) and recipients with more than two children ([>2 Children]). This categorization is inferred from the amount of the monthly CTC payments received. We estimate the model separately for both enhanced CTC recipients and advance-only recipients, as in Table 3.

The results are presented in panel C of Table 3. The results indicate a stronger effect for recipients with more children, as expected. The difference in coefficients on the interaction terms of interest is statistically significant in all models, at the 1% level. Enhanced recipients with more than two children increase spending by \$172, almost twice the increase for enhanced recipients with one or two children, \$87. Interestingly, advance-only recipients with more than two children increase spending by \$108 dollars in the week after payment, relative to the prior week, whereas advance-only recipients with one or two children do not significantly increase spending. This suggests that the overall

increase we observe for advance-only recipients is driven by those with larger families, and thus larger payment amounts.

Having established that the CTC payments significantly influence overall consumption patterns, we next examine categories that align with the intended purposes of the CTC, as articulated by policymakers. Public statements and legislative debates highlighted the CTC's expected role in alleviating pressures on essential spending. For example, Vice President Harris emphasized the burden of childcare costs and the necessity for consistent financial support for groceries and gas, highlighting how the CTC helps families meet these basic needs. President Biden and Senator Markey both stated that the CTC supports expenditures on healthcare, education, and food.¹⁸ These categories—groceries, other food/dining, healthcare, education, and gas—were emphasized as critical in providing immediate relief to families and enabling them to meet basic needs, a central goal of the regulation.

Table 4 presents our analysis of consumption patterns across selected essential categories that align with the legislative goals of the CTC. We estimate separate regressions, similar to Equations (1) and (2), but focusing on disaggregated spending within relevant categories—namely, groceries, other food/dining, education, healthcare, and gas. To also capture increased cash spending, which could fall within these categories or extend to other categories, we include cash withdrawals. Our findings indicate that enhanced recipients significantly increased their consumption in all these areas. Advance-only recipients did not significantly increase spending in healthcare, but increased spending in all other categories examined. Comparing the two groups, although both enhanced and advance-only recipients showed similar increases in education spending, enhanced recipients exhibited a more pronounced increase in spending on groceries, other food/dining, healthcare, and gas. Enhanced recipients also demonstrated a stronger uptick in cash withdrawals, suggesting increased cash spending, which could fall within these same categories, or extend to other categories.¹⁹

Together, the results in this section indicate that monthly CTC payments drive an immediate and significant increase in consumption. This increase is stronger for enhanced recipients than advance-only ones, and for families with more children. These results collectively suggest that the short-term spending responses around CTC payments are likely driven by those payments and not other confounding effects, and provide insights into how recipients use the payments they receive.

4.2. Liquidity Constraints

In this section, we examine whether CTC payments impact liquidity constraints, to shed light on whether

Table 4. Consumption by Category

(1) DID estimates for enhanced sample	(2) DID estimates for advance-only sample	(3) Difference between enhanced and advance-only
	Groceries: DV = $\text{LnGroceries}\$Adj$	
0.118*** (41.437)	0.025*** (3.718)	0.093*** (13.108)
	Other food/dining: DV = $\text{LnOther_Food_Dining}\Adj	
0.130*** (52.718)	0.029*** (4.812)	0.102*** (16.232)
	Education: DV = $\text{LnEducation}\$Adj$	
0.013*** (8.327)	0.011** (2.008)	0.002 (0.388)
	Healthcare: DV = $\text{LnHealthcare}\$Adj$	
0.058*** (19.276)	0.004 (0.560)	0.054*** (6.494)
	Gas: DV = $\text{LnGas}\$Adj$	
0.092*** (29.005)	0.031*** (3.905)	0.061*** (7.221)
	Cash: DV = $\text{LnCash}\$Adj$	
0.113*** (35.470)	0.025*** (2.989)	0.088*** (10.170)

Notes. This table reports the impact of CTC expansion on weekly consumption in six categories. Column (1) uses the matched enhanced CTC sample, and column (2) uses the matched advance-only CTC sample. The sample includes observations from July through September 2021 with one week before the receipt of a monthly CTC payment and one week after. Each cell represents one regression, and we only report the DID estimate for brevity. The dependent variables (DV) are logged weekly consumption in each category adjusted by its value in the same week in 2019. In column (3), we report the result of tests for the coefficient difference between the two samples using two-tailed *t*-tests, based upon stacked regressions which estimate effects for both groups jointly. Please refer to Appendix A for detailed variable definitions. We control for individual fixed effects and state-month fixed effects. *T*-statistics based on standard errors clustered by individual are reported in parentheses. ***, **, and * stand for statistical significance at the 1%, 5%, and 10% level, respectively.

monthly CTC payments help families to manage their income-expense timing. To gauge this, we examine overdraft charges, an indicator that the individual is unable to cover their expenses; payday loans, an indicator that the individual resorts to high-cost loans to pay some bills; and gig work, an indicator that the individual takes side gigs to help make ends meet. Although these events are infrequent, occurring in 1%–2% of recipient-months for enhanced recipients, and 1% or fewer of recipient-months for advance-only recipients, they serve to indicate that individuals have essentially run out of funds (overdraft or payday loan use), or are taking on additional work to earn additional funds (gig work). These events are an indicator of financial hardship. Thus, examining these outcomes sheds light on the extent to which CTC payments reduce financial hardship for recipients.

Table 5 presents the results of the DID analyses. Columns (1)–(2) focus on overdrafts, and they show that both enhanced and advance-only recipients experience a drop in overdraft charges in the week after CTC payments, relative to the prior week.

Although the coefficient size is small, relative to the low rate of overdraft charges in a short window (about 0.4%–1% of the sample, as shown in Table 2), the estimates suggest that 35.4% fewer enhanced recipients

experience overdrafts in the week after payments than the week before ($= 0.0034/0.0096 \times 100\%$), and 15.8% fewer advance-only recipients experience overdrafts ($= 0.0006/0.0038 \times 100\%$). Thus, it appears that monthly payments significantly support both groups in smoothing their consumption without overdrawing their accounts. The reduction in overdraft instances is significantly stronger for enhanced recipients than for advance-only recipients, likely because of both increased payment amounts and a higher risk of binding liquidity constraints for enhanced recipients.

Columns (3)–(4) focus on the use of payday loans. Using payday loans is rare for either group, consistent with payday loans serving as a loan of last resort for most people. However, there is still a considerable rate of usage in our sample, at 0.7% per week for enhanced recipients, and 0.1% for advance-only recipients. We find that only enhanced recipients significantly reduce the use of payday loans in the week after receiving CTC payments. Economically, the estimates in column (3) suggest a reduction of about 9.9% ($= 0.0007/0.0071 \times 100\%$) in the occurrence of new payday loans. Given the extremely high cost of payday loans and the risk that they spur a debt spiral, the 9.9% reduction may be particularly significant from a welfare perspective. There is no significant reduction in the use of payday

Table 5. Liquidity Constraints

Variables	(1) Overdraft(D)Adj	(2) Payday_Loan(D)Adj	(3) Gig_Work(D)Adj	(4)	(5)	(6)
Week[+1]	−0.0014*** (−8.945)	−0.0004 (−1.414)	−0.0004*** (−4.668)	0.0000 (0.093)	−0.0010*** (−7.979)	−0.0010*** (−3.461)
Week[+1] × Enhanced_CTC	−0.0034*** (−17.309)		−0.0007*** (−6.558)		−0.0003** (−2.001)	
Week[+1] × Advance-Only_CTC		−0.0006* (−1.757)		−0.0001 (−0.961)		0.0002 (0.560)
Difference in the DID estimates for enhanced and advance-only samples	−0.0028*** (−7.338)		−0.006*** (−3.354)		−0.0005 (−1.36)	
State-month FE	Y	Y	Y	Y	Y	Y
Individual FE	Y	Y	Y	Y	Y	Y
Observations	10,496,784	1,451,796	10,496,784	1,451,796	10,496,784	1,451,796
Adj R ²	0.174	0.148	0.403	0.394	0.409	0.369

Notes. This table reports the impact of CTC expansion on liquidity constraints. Columns (1), (3), and (5) use a matched enhanced CTC sample (874,732 enhanced CTC recipients and their matches), and columns (2), (4), and (6) use a matched advance-only CTC sample (120,983 advance-only recipients and their matches). The sample includes observations from July through September 2021 with one week before the receipt of a monthly CTC payment and one week after. The dependent variables in columns (1)–(2) are a dummy variable indicating whether overdraft charges occur to the individual or not. The dependent variables in columns (3)–(4) are a dummy variable indicating whether the individual borrows from a payday loan app or not. The dependent variables in columns (5)–(6) are a dummy variable indicating whether the individual takes gig jobs or not. All dependent variables are benchmarked against the same week in 2019. The row labeled “Difference in the DID estimates for enhanced and advance-only samples” reports results for tests of the coefficient difference between the enhanced CTC and advance-only CTC samples using two-tailed tests, based upon stacked regressions which estimate effects for both groups jointly. Please refer to Appendix A for detailed variable definitions. We control for individual fixed effects and state-month fixed effects. *T*-statistics based on standard errors clustered by individual are reported in parentheses.

***, **, and * stand for statistical significance at the 1%, 5%, and 10% level, respectively.

loans for advance-only recipients, demonstrating a significant difference in this respect between the two groups. Thus, it is unclear if the benefit to enhanced recipients is due to monthly timing, increased payment amounts, or their combination.

Finally, columns (5)–(6) focus on gig work. Consistent with the results from overdrafts and payday loans, we again find a significant reduction in gig work participation among enhanced recipients in the week after receiving CTC payments; however, the effect is small. Economically, the estimate suggests that enhanced recipients reduce gig work participation by 1.8% ($= 0.0003/0.0164 \times 100\%$). As with payday loans, we find no significant change for advance-only recipients. The difference in the impact on gig work participation between the two groups of recipients is insignificant. The results are consistent with CTC payments reducing liquidity constraints for enhanced recipients, such that they rely less on gig work for extra money to meet their daily needs in the week following a payment. Overall, the results from overdrafts, payday loans, and gig work are consistent with the expectation that monthly CTC payments reduce liquidity constraints and that they are generally more impactful for low-income recipients, and with increased payment amounts. We build on these results in additional analyses, reported in Section 4.3.3. As we discuss in that section, the timing of payments—monthly versus annual—can impact liquidity dynamics. We examine payment timing more directly in that section.

4.3. Policy Implications

In this section, we conduct additional analyses to address specific policy-relevant questions. First, we address several questions related to different income brackets. Second, we compare monthly CTC payments with annual tax refunds.

4.3.1. Full Refundability. The CTC is traditionally, and primarily, a tax credit, which can be refundable or not. To explain the potential impact of refundability, we provide a hypothetical example contrasting nonrefundability with full refundability. Consider a family which would otherwise owe \$10,000 in income taxes in a given year. If they had a single child with an annual CTC of \$2,000, they would instead owe \$8,000 in taxes. Thus, they would get a CTC benefit of \$2,000. However, if a family owed only \$1,000 in taxes, because of having a sufficiently low income, they would not receive the full \$2,000 benefit under nonrefundability. Specifically, under nonrefundability, this family would receive a total of \$1,000 in benefit, reducing their income tax to \$0, but not receiving any additional benefit. Under full refundability, this family would pay \$0 income tax and receive a refund of \$1,000, and thus receive the full \$2,000 CTC benefit.

Refundability is an important issue for CTC policy, given the child-poverty-reduction goals of CTC payments. Under nonrefundability, lower-income households receive lower payments/benefits. Some argue

that because of this, CTC funds were not reaching the children who needed them most under nonrefundability. The CTC was nonrefundable when it was first introduced in 1997, but it was modified to be partially refundable starting with the Economic Growth and Tax Relief Reconciliation Act (EGTRRA) of 2001, to better reach children living in poor households. The CTC expansion under the ARPA of 2021 went significantly further and made the CTC fully refundable. With the 2021 CTC expansion, the lower-income family described above would receive the full \$2,000 benefit. In this sense, the expanded CTC could be considered a government transfer policy, rather than a pure tax credit.

An important policy-relevant question is how families affected by full refundability used the increased funds. This is relevant both from a child-poverty-reduction standpoint and to specifically address the question of the impact of refundability. We face certain data limitations which affect our ability to fully evaluate the impact of full refundability. In particular, as discussed in Section 2.1, our data do not include unbanked individuals or individuals who use primarily cash for income/wages and payments. Thus, our data may underrepresent lower-income recipients. However, our data still capture over 100,000 CTC recipients with estimated annual income below \$45,000, a demographic likely to benefit from full refundability.²⁰ Thus, to provide policy-relevant insights, we focus on enhanced recipients and augment Model (1) by including an additional interaction term for an indicator of estimated annual income below \$45,000 ($<45K_Income$).

Results are presented in Table 6, panel A. The results indicate that these recipients increase their consumption by the largest proportion, with a significantly positive coefficient on the triple interaction term $Week[+1] \times Enhanced_CTC \times <45K_Income$. In terms of dollar magnitudes, recipients with incomes below \$45,000 increase spending by roughly \$156 from the week before to the week after receiving a monthly CTC payment. This is higher than the spending increase of roughly \$95 for enhanced recipients with incomes higher than \$45,000, and an increase of roughly \$22 for advance-only recipients, as reported in Table 3. The increase in postweek spending of \$156 translates to roughly 39% of the average monthly CTC payment received by these recipients (\$399.66). This suggests that a large portion of monthly CTC payments to lower-income recipients, who are likely benefiting from full refundability under the plan, are used for immediate consumption, and highlights the importance of full refundability for the highest-need recipients.

4.3.2. Phase-Out Level. A second policy-relevant question is whether the income threshold for phase-out of payments is appropriate. In particular, the TCJA of 2017 increased the income at which CTC payments

phase out. As indicated in Figure 1, the TCJA raised the phase-out income level from \$110,000 for a married couple filing jointly (green line) to \$400,000 (red/blue line). It also raised the phase-out from \$75,000 to \$200,000 for a parent filing singly or as head of household. Although the 2021 expansion did not change the phase-out thresholds relative to TCJA, the switch to monthly payments allows us to more precisely measure the consumption response of higher-income individuals to CTC payments. Given that the TCJA changes are scheduled to expire in 2025 and phase-out choice impacts the cost and benefits of any proposed CTC policy, Congress will need to consider what phase-out thresholds to utilize in any future legislation.

To provide evidence on current and possible phase-out levels, we conduct two analyses. First, we focus on advance-only recipients to examine variation in short-term spending response within this group. The median advance-only recipient in our data is estimated to have an income tier of above \$150,000 (Table 1, panel B). Thus, we augment Model (2) with an additional interaction term for an indicator of annual income below \$150,000 ($<150K_Income$), in other words, with lower income than the median advance-only recipient. This allows us to examine variation among advance-only recipients in their use of CTC payments and provide evidence to inform discussions of alternative phase-out income levels.

The results are presented in panel B of Table 6, and indicate that the higher-income advance-only recipients, those with estimated income above \$150,000, do not change their consumption after receiving CTC payments, as the coefficient on $Week[+1] \times Advance_only_CTC$ is statistically insignificant. The significant consumption response for advance-only CTC recipients documented in Table 3 is primarily driven by the subgroup with an estimated income below \$150,000, as indicated by the positive coefficient estimates for this group. As indicated in column (2), the average advance-only recipient with an estimated income below \$150,000 spends \$33.60 of the CTC payment in the week after receipt. These results provide suggestive evidence that CTC payments do not significantly affect the consumption behavior of higher-income advance-only recipients. Thus, policy setters may want to consider a lower income threshold for the phase-out of CTC payments.²¹

To provide additional evidence regarding the threshold for phase-out of CTC payments, we explicitly examine phase-out CTC recipients with monthly CTC payments below \$166. These high-income recipients receive a smaller payment than advance-only recipients (see Figure 1). We apply the same matching algorithm as used for our main sample construction to identify a nonrecipient match for each phase-out recipient and compare their short-term spending in the weeks before and after the receipt of CTC payments. As shown in

Table 6. Income Thresholds for CTC

Variables	(1) LnTot_Cons\$Adj	(2) Tot_Cons\$Adj
Panel A: Enhanced recipients: full refundability threshold (\$45K annual income)		
Week[+1]	−0.056*** (−28.536)	−67.041*** (−36.089)
Week[+1] × Enhanced_CTC	0.103*** (42.431)	95.300*** (41.969)
Week[+1] × <45K_Income	0.004 (0.782)	53.400*** (19.993)
Week[+1] × Enhanced_CTC × <45K_Income	0.458*** (63.862)	60.274*** (17.473)
DID estimates when <45K_Income = 1	0.561*** (83.08)	155.575*** (59.91)
State-month FE	Y	Y
Individual FE	Y	Y
Observations	10,496,782	10,496,784
Adj R ²	0.412	0.182
Panel B: Advance-only recipients: median annual income \$150K		
Week[+1]	−0.064*** (−10.740)	−106.125*** (−13.814)
Week[+1] × Advance-only_CTC	0.013 (1.631)	16.371 (1.637)
Week[+1] × <150K_Income	0.022** (2.256)	62.152*** (6.267)
Week[+1] × Advance-only_CTC × <150K_Income	0.034** (2.556)	17.234 (1.296)
DID estimates when <150K_Income = 1	0.046*** (4.33)	33.606*** (3.84)
State-month FE	Y	Y
Individual FE	Y	Y
Observations	1,451,796	1,451,796
Adj R ²	0.325	0.150
Panel C: Phase-out recipients		
Week[+1]	−0.065*** (−3.173)	−104.314*** (−3.334)
Week[+1] × Phase-out_CTC	−0.019 (−0.628)	−13.854 (−0.304)
State-month FE	Y	Y
Individual FE	Y	Y
Observations	58,064	58,064
Adj R ²	0.326	0.112

Notes. This table investigates different income thresholds that determines the 2021 CTC payouts. The dependent variables are adjusted logged or unlogged weekly total consumption, as indicated in the table header. Panel A uses the matched sample for enhanced CTC recipients. The partition variable, *<45K_Income*, is a dummy indicator equal to one when the individual is in income tiers 1 and 2 (corresponding to annual income below \$45,000) and zero otherwise. Panel B uses the matched sample for advance-only CTC recipients. The partition variable, *<150K_Income*, is a dummy indicator equal to one when the individual is in an income tier below 7 (7 is the median income tier for advance-only recipients, corresponding to an annual income of \$150,000 or above). Panel C uses the same algorithms to construct a matched sample for phase-out recipients. We identify a clean set of phase-out recipients based on monthly CTC payments below \$166. In panels A and B, the rows labeled “DID estimates when *<45K_Income* = 1” and “DID estimates when *<150K_Income* = 1” report the total DID effects for the given recipient and income samples (e.g., the sum of coefficients for Week[+1] × Recipient and Week[+1] × Recipient × Income_Category_Indicator). Please refer to Appendix A for detailed variable definitions. We control for individual fixed effects and state-month fixed effects. *T*-statistics based on standard errors clustered by individual are reported in parentheses.

***, **, and * stand for statistical significance at the 1%, 5%, and 10% level, respectively.

panel C of Table 6, we fail to detect significant spending responses by these phase-out CTC recipients. These results complement those in panel B and seem to suggest that a lower phase-out threshold might be an effective way to lower costs without compromising the

effectiveness of the CTC in addressing the needs of financially constrained families.

4.3.3. Monthly vs. Once-Yearly Payments. Another important policy-relevant question is whether receiving

tax credits on a monthly basis throughout the year has a differential impact on recipients than receiving the credit as part of an annual tax refund. Although our main analyses address the overall effect of monthly payments, they do not explicitly compare monthly and annual payments. Prior empirical research provides evidence that taxpayers spend a disproportionate amount of tax refunds soon after receiving the refund, failing to use refunds to save for the future and smooth consumption (e.g., Souleles 1999, Baugh et al. 2021). Behavioral research has also examined the impact of payment timing and structure on saving and spending patterns, though with unclear implications for the CTC setting (LaMothe et al. 2024). We expect that monthly periodic payments may better support recipients' saving and smoothing behaviors, in comparison with an equivalent one-time annual payment.

To examine the effects of monthly payments, we first focus on income-smoothing effects. One of the potential benefits of monthly payments, as opposed to annual ones, is that monthly payments can facilitate financial planning and ease liquidity constraints throughout the year. To examine whether this is the case, we focus on the occurrence of overdrafts, use of payday borrowing, and gig work participation, all of which indicate binding liquidity constraints and a need for supplemental tools to mitigate these constraints and satisfy consumption needs. In addition, we explicitly examine whether payments support consumption smoothing by examining consumption volatility.

We compare these measures during the three months of CTC payments (July–September 2021) with the three months following 2020 tax refunds (for refunds received during the January–June 2021 period). Although the refunds are not specifically CTC payments, they provide a benchmark to evaluate the effects of an annual refund/payment. The sample for this analysis is restricted to 468,343 enhanced recipients and 37,235 advance-only recipients who receive a refund in the first half of 2021 (i.e., excluding CTC recipients with tax refunds received later or with tax payments). The regression specification is as follows:

$$\begin{aligned} Outcome = & \gamma_0 + \gamma_1 CTC(D) + \gamma_2 LnAmount \\ & + Income_Tier_FE + StateFE + \epsilon, \end{aligned} \quad (3)$$

where each sample individual appears twice in the sample, with one observation representing the three months following the first CTC payment, and one observation representing the three months following the 2020 tax refund. The model includes state fixed effects and income tier fixed effects.²² The dependent variables are an indicator for overdrafts, an indicator for payday loan use, an indicator for gig work participation, and consumption volatility during the three-month window following the respective CTC or refund

payments, benchmarked against the same period in 2019. We define overdrafts, payday loans, and gig work participation similarly to Table 5 but, however, using a three-month window following CTC/refund payment, instead of one week. Consumption volatility is defined as the standard deviation of daily total consumptions during the three months following the respective payments. We control for the total amount of payments. Thus, the coefficient estimate on the key variable of interest, CTC indicator ($CTC(D)$), captures the incremental effect of monthly payments (as opposed to tax refunds), controlling for the total magnitude of those payments.

Table 7, panel A, reports summary statistics for the sample and variables used in this analysis. The average 2020 tax refund is roughly three times the amount of the average three-month CTC payment, suggesting that the amounts are economically comparable. However, it is possible that for some individuals, the refund amount is particularly low or high, compared with CTC payments. In untabulated analysis, we restrict the subsample to observations for which the refund amount ranges from 1 to 12 times the monthly CTC payment—in other words, between one month and one year of monthly CTC payments. Results are qualitatively similar.

We report the regression results in Table 7, panel B. Columns (1)–(4) compare tax refunds with monthly CTC payments for enhanced recipients. Focusing first on the three measures of liquidity constraints, the negative coefficients on $CTC(D)$ indicate a significant reduction in all three—the occurrence of overdrafts, the use of payday lending, and the use of gig work—during the period with monthly CTC payments, for enhanced recipients. This result provides suggestive evidence that, compared with a lump-sum payment as part of tax refunds, paying out the CTC benefit throughout the year enables families to better manage their finances and reduce liquidity constraints.²³

Column (4) compares the impact of once-yearly tax refunds with monthly CTC payments on consumption volatility for enhanced recipients. We find a significant reduction in consumption volatility during the three months of monthly CTC payments, relative to the three months following tax refunds. This is consistent with monthly payments supporting consumption smoothing for enhanced recipients.

Results reported in columns (5) through (8) show less consistent evidence for advance-only recipients. We find reductions in overdrafts and gig work, but not the use of payday lending or consumption volatility. It may be that advance-only recipients are less liquidity constrained in general, such that payment timing does not impact liquidity constraints and consumption smoothing as strongly for advance-only recipients.²⁴

An alternative way to examine the effect of annual versus monthly payments on consumption is to

Table 7. Comparison of CTC Monthly Payments and Annual Tax Refunds

Panel A: Summary statistics for the CTC vs. refund comparison sample								
Variables	Enhanced subsample				Advance-only subsample			
	Refund N = 468,343		CTC N = 468,343		Refund N = 37,235		CTC N = 37,235	
	Mean	S.D.	Mean	S.D.	Mean	S.D.	Mean	S.D.
\$Refund	4,638.457	2,956.880	N/A	N/A	2,816.792	2,333.724	N/A	N/A
\$CTC3M	N/A	N/A	1,484.454	713.967	N/A	N/A	904.389	408.906
Overdraft(D)3M_Adj	−0.031	0.276	−0.037	0.296	−0.013	0.219	−0.018	0.229
Payday_Loan(D)3M_Adj	0.000	0.114	−0.001	0.121	0.000	0.073	0.000	0.075
Gig_Work(D)3M_Adj	0.017	0.286	0.017	0.289	0.019	0.262	0.016	0.258
Cons_Vol\$3M_Adj	147.2	521.1	86.2	500.7	127.1	783.8	105.8	761.0
Panel B: Regression analyses comparing CTC monthly payments and annual tax refunds								
Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Enhanced subsample				Advance-only subsample			
	Overdraft(D) 3M_Adj	Payday_Loan(D) 3M_Adj	Gig_Work(D) 3M_Adj	Cons_Vol\$ 3M_Adj	Overdraft(D) 3M_Adj	Payday_Loan(D) 3M_Adj	Gig_Work(D) 3M_Adj	Cons_Vol\$ 3M_Adj
CTC(D)	−0.004*** (−5.566)	−0.001*** (−4.072)	−0.003*** (−3.926)	−35.681*** (−27.240)	−0.006*** (−2.994)	−0.001 (−1.473)	−0.010*** (−4.533)	−9.867 (−1.558)
LnAmount	0.002*** (4.044)	−0.000*** (−2.650)	−0.003*** (−7.093)	25.938*** (32.697)	−0.001 (−1.133)	−0.000 (−1.278)	−0.008*** (−6.409)	14.041*** (3.898)
State FE	Y	Y	Y	Y	Y	Y	Y	Y
Income tier FE	Y	Y	Y	Y	Y	Y	Y	Y
Observations	936,686	936,686	936,686	936,686	74,470	74,470	74,470	74,470
Adj R ²	0.008	0.000	0.000	0.028	0.005	0.001	0.001	0.010

Notes. This table compares the impact of monthly CTC payments with annual tax refunds. The sample includes individuals who receive CTC payments during our sample period (July–September 2021) and a tax refund from January to June 2021 (i.e., each individual appears in the sample twice, once for the three months of CTC payments and the other for the three months after the refund). Panel A reports the summary statistics for the sample, and Panel B reports the regression results. Columns (1)–(4) compare enhanced CTC payments with refunds whereas columns (5)–(8) compare advance-only CTC payments with refunds. The dependent variables are indicators for overdrafts, payday loan takeouts, gig work participation, and volatilities of daily total consumption over the three months following the first CTC payment or the tax refund, as indicated in the heads. The key variable of interest is the dummy variable, *CTC(D)*, which is equal to one for CTC payments and zero for refunds. We control for the amount of payment (*LnAmount*, logged amount of three months' CTC payments or the tax refunds received). State fixed effects and income tier fixed effects are included in each regression. *T*-statistics based on robust standard errors are reported in parentheses.

***, **, and * stand for statistical significance at the 1%, 5%, and 10% level, respectively.

examine the time trend in consumption during the relevant postpayment periods. Given prior evidence that taxpayers spend a disproportionate amount of tax refunds soon after receiving the refund (Souleles 1999, Baugh et al. 2021), we expect higher spending immediately after a tax refund than after the first monthly CTC payment. However, postrefund spending should drop significantly as time passes, whereas post-CTC spending should be more stable, given subsequent monthly payments. Figure 5 presents weekly spending following tax refunds and the first monthly CTC payment, similar to Baugh et al. (2021). We find the expected consumption patterns for enhanced recipients, but find that payment timing makes little difference for spending patterns of advance-only recipients.

In untabulated analyses, we estimate two separate regression models that approximate a linear time trend in consumption in the three months following tax refunds versus the three months following the first monthly CTC payment. We find a significantly stronger

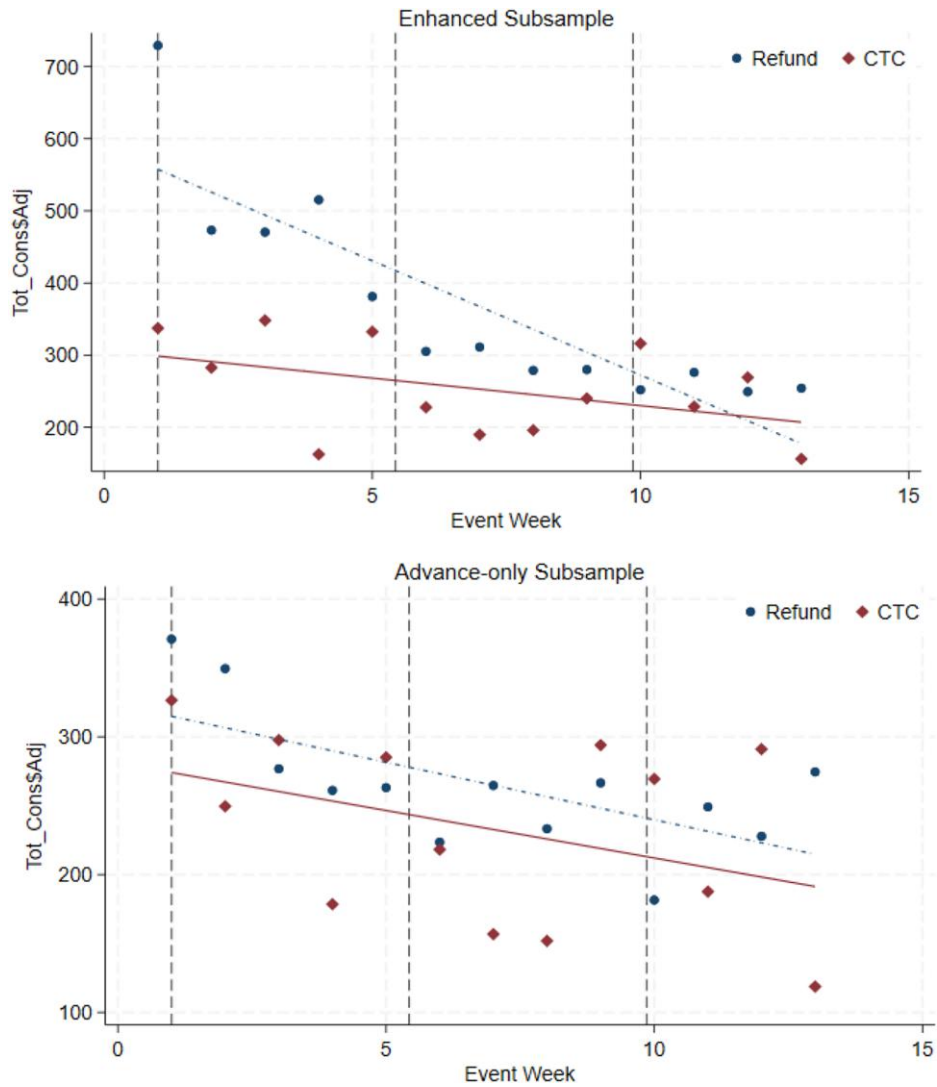
decline in spending over the weeks following a tax refund for enhanced recipients, compared with the trend during the post-CTC payment period. However, we do not find a consistent difference between postrefund and CTC-payment-period spending patterns for advance-only recipients.²⁵

Together, these results provide consistent evidence that monthly payments assist in consumption smoothing and reduce the occurrence of negative liquidity constraint events for enhanced recipients, even for similar cumulative payment amounts. This is relevant as regulators consider payment timing for future CTC legislation.

4.4. Financial Position

The results reported above show that lower-income recipients of CTC payments spend a large portion of CTC payments on immediate consumption. Further, the results above show a decrease in liquidity constraints. Yet, it remains unclear whether recipients are

Figure 5. (Color online) Weekly Consumption Following Refunds and CTC Payments



Notes. This figure plots the average weekly consumption for 13 weeks (equivalent to three months) following the receipt of refunds (blue dots) and the first monthly CTC payment (red diamonds). To facilitate the comparison of the consumption pattern, we add a linear fitted line. We also add three dashed vertical lines to indicate the approximate arrivals of CTC payments, with the first arrival in week 1 and the second and third estimated on 31 and 62 days after the initial CTC payment.

able to use CTC payments to improve their overall financial positions—leading to higher savings or paying down debt, thereby offering longer-term benefits. In this section, we examine whether CTC recipients increase their payments toward financial positions—savings, investments, debt (e.g., personal loans, student loans, auto loans, and mortgages), and credit cards—providing additional descriptive evidence on how recipients use CTC payments.

Table 8 presents the results of DID analyses for financial positions. Both enhanced and advance-only recipients increase payments toward all four of these categories, relative to matches, suggesting that CTC payments allow recipients to improve their overall financial positions.²⁶ Enhanced recipients proportionally increase

contributions toward savings and payments toward credit cards more than advance-only recipients. These short-term increases in payments toward savings, investments, and debt reduction may provide a longer-term benefit to recipients, strengthening their financial positions going forward.

5. Conclusion

We examine the realized effects of the CTC expansion which occurred as part of the American Rescue Plan Act of 2021. Consistent with the goals of the 2021 CTC reform to support families, our results indicate that recipients increase their total consumption in a wide range of categories, including groceries, healthcare, gas, and

Table 8. Supplemental Analysis: Short-Term Financial Position

Variables	(1) Savings LnSaving\$Adj	(2)	(3) Investment LnInvest\$Adj	(4)	(5) Debt repayments LnLoan\$Adj	(6)	(7) Credit card payments LnCC_Payment\$Adj	(8)
Week[+1]	−0.000 (−0.182)	0.003 (1.171)	0.021*** (12.919)	0.036*** (6.604)	−0.084*** (−18.337)	−0.173*** (−13.841)	−0.060*** (−13.791)	−0.122*** (−10.177)
Week[+1] × Enhanced_CTC	0.043*** (34.124)		0.019*** (9.440)		0.069*** (11.954)		0.256*** (46.689)	
Week[+1] × Advance-Only_CTC		0.034*** (7.336)		0.027*** (3.536)		0.053*** (3.223)		0.080*** (5.001)
Difference in the DID estimates for enhanced and advance-only	0.009** (2.016)		−0.008 (−1.058)		0.016 (0.934)		0.175*** (10.567)	
State-month FE	Y	Y	Y	Y	Y	Y	Y	Y
Individual FE	Y	Y	Y	Y	Y	Y	Y	Y
Observations	10,496,784	1,451,796	10,496,784	1,451,796	10,496,782	1,451,796	10,496,784	1,451,796
Adj R ²	0.272	0.227	0.285	0.247	0.199	0.189	0.141	0.096

Notes. This table reports the impact of the CTC expansion on different proxies for financial position. Columns (1), (3), (5), and (7) use the matched enhanced CTC sample, and columns (2), (4), (6), and (8) use the matched advance-only CTC sample. The sample includes observations from July through September 2021 with one week before the receipt of a monthly CTC payment and one week after. The dependent variables are saving (logged amount of the money transferred to savings accounts) in columns (1) and (2), investment (logged amount of the money transferred to brokerage accounts) in columns (3) and (4), debt repayments (logged amount of money outflows in the categories of “Loans” and “Mortgages”) in columns (5) and (6), and credit card payments (logged amount of money outflows in the category of “Credit Card Payments”) in columns (7) and (8). All dependent variables are adjusted by their value in the same week in 2019. The row “Difference in DID estimates ...” reports results for two-tailed *t*-tests of the coefficient differences between the enhanced CTC and advance-only CTC samples, based upon stacked regressions which estimate effects for both groups jointly. Please refer to Appendix A for detailed variable definitions. We control for individual fixed effects and state-month fixed effects. *T*-statistics based on standard errors clustered by individual are reported in parentheses.

***, **, and * stand for statistical significance at the 1%, 5%, and 10% level, respectively.

education. These consumption effects are particularly pronounced for families with lower incomes. Enhanced recipients, whose total payment amount increased because of the legislation, spend on average \$103 more, 22% of their monthly payment, in the week following payment receipt compared with the week prior. Advance-only recipients also increase consumption significantly, but at a lower rate. Advance-only recipients increase spending by \$22 in the week following receipt, 8% of their average monthly payment.

We provide more direct evidence on the liquidity provision benefits of monthly CTC payments by examining changes in liquidity constraint-related events: overdrafts, the use of high-interest payday loans, and the use of gig work to alleviate liquidity shortfalls and smooth consumption. Our findings indicate a reduction in all three categories for enhanced recipients. The 9.9% reduction in payday loans is particularly notable from a welfare perspective given the potential for high-interest payday loans to lead to a debt cycle.

Together, these results suggest that both monthly payments and increased payment amounts facilitate immediate consumption and ease liquidity constraints for recipients. We conduct several additional analyses to speak to specific policy questions. The lowest-income recipients, who likely benefit from full refundability under the 2021 plan, spend at an even higher rate than other enhanced recipients. They spend roughly 39% of their monthly CTC payments in the week following the receipt. In contrast, the top half of

advance-only recipients, determined by income, have no significant consumption increase. Similarly, individuals in the (even higher income) phase-out range also do not increase consumption. These findings suggest that if the primary goal of the CTC is to facilitate consumption, a viable way to reduce CTC costs would be to reevaluate the income level at which benefits start to phase out.

Finally, we directly compare monthly and once-yearly payments, using tax refunds as a basis of comparison. Our findings indicate that monthly payments drive an additional reduction in liquidity constraints, controlling for the amount of payment an individual receives. Monthly payments facilitate consumption smoothing, reducing the volatility of weekly consumption and leading to more stable consumption over time. These results shed additional light on the potential advantages of opting for monthly payments over annual ones.

Our results complement concurrent studies on the CTC, which examine employment effects and/or self-reported hardship measures. By analyzing transaction data from a substantial sample of Americans, our study offers the first empirical evidence, derived from actual transactions, on the realized impacts of the 2021 CTC expansion. This evidence can significantly contribute to the continuing debate surrounding the CTC. Although our data do not include unbanked individuals, they nevertheless span a wide cross-section of the American population, both geographically and in terms of

income. The insights gleaned from our data and analyses provide valuable information on the impact of the CTC on the families it aims to support and can be used to inform current and future policy discussions.

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Appendix A. Variable Definitions

For all variables below, a week refers to seven consecutive days, where the week can start from any day of the week and vary across individuals, depending on the timing of CTC payments. A prefix of “Ln” indicates that we take the natural logarithm of one plus the variable’s value. A suffix of “Adj” means the variable is adjusted by its value in the same week (same seven calendar days) in 2019.

Variable	Definition
Variables for the main sample	
Tot_Cons\$	Weekly total consumption (identified by debit transactions in the categories other than “Loans,” “Mortgages,” “Credit Card Payments,” “Savings,” “Retirement Contributions,” “Securities Trades,” “Taxes,” and bank transfers).
Groceries\$	Weekly grocery spending (identified by debit transactions in the category of “Groceries”). Please refer to Appendix B for top merchants in this category.
Other_Food_Dining\$	Weekly spending on other dining establishments, such as fast food and restaurants (identified by debit transactions in the category of “Restaurants”). Please refer to Appendix B for top merchants in this category.
Education\$	Weekly education spending (identified by debit transactions in the category of “Education”). Please refer to Appendix B for top merchants in this category.
Healthcare\$	Weekly healthcare spending (identified by debit transactions in the category of “Healthcare/ Medical”). Please refer to Appendix B for top merchants in this category.
Gas\$	Weekly spending on gas (identified by debit transactions in the category of “Automotive/ Fuel”). Please refer to Appendix B for top merchants in this category.
Cash\$	Weekly cash withdrawals (identified by debit transactions in the category of “ATM/Cash Withdrawals”).
Overdraft(D)	A dummy variable indicating the existence of overdraft/nonsufficient charges to the bank account during a week, identified using regular expression matching based on keywords such as “insufficient funds,” “overdraft,” and “fee”/“charge”/“chrg”/“CHG,” “OD CHG,” and “OD FEE.” The full list of keywords is available upon request.
Payday_Loan(D)	Dummy variable, equal to one if nonzero money flows into the bank account from payday lending or cash advance apps (Brigit, Dave, Earin, Empower, FloatMe, Clover, MoneyLion, Possible, Instacash) during a week.
Gig_Work(D)	Dummy variable, equal to one if receiving money from nearly 50 gig platforms such as Uber and Upwork at least once during a week. The full list of gig platforms is available upon request.
Saving\$	Money put into savings accounts (identified by debit transactions in the category of “Savings”) during a week.
Invest\$	Money put into brokerage accounts (identified by debit transactions to 129 brokerages such as Robinhood and TD Ameritrade or debit transactions in the categories of “Securities Trades” when transaction counterparties are missing) during a week.
Loan\$	Debt payments (identified by debit transactions in the category of “Loans” and “Mortgages”) during a week.
CC_Payment\$	Credit card payments (identified by debit transactions in the category of “Credit Card Payments”) during a week.
Week[+1]	Dummy variable, equal to one for the week starting on the day of CTC payments and zero for the week before the payments.
Enhanced_CTC	Dummy variable, equal to one for CTC recipients whose CTC payments are multiples or combinations of \$250 or \$300 per month.
Advance-Only_CTC	Dummy variable, equal to one for CTC recipients whose CTC payments are multiples or combinations of \$166 or \$167 per month.
Phase-out_CTC	Dummy variable, equal to one for clearly identified CTC recipients whose CTC payments are below \$166 per month.
\$CTC	Dollar amount of CTC payment per month.
#Children	Number of children, inferred from the dollar amount of monthly CTC payments.

Appendix. (Continued)

Variable	Definition
Income Tier	Income tier as of June 2021, ranging from 1 to 7 as determined by the data provider based on prior salary, other income, and spending. The meaning of the variable value is as follows: 1, \$0–25K; 2, \$25–45K; 3, \$45–60K; 4, \$60–75K; 5, \$75–100K; 6, \$100–150K; 7, \$150K+.
<45KIncome	Dummy variable, equal to one for when an individual’s income tier as of June 2021 is below 45K (income tier in 1 and 2).
<150KIncome	Dummy variable, equal to one for when an individual’s income tier as of June 2021 is below 150K (income tier from 1 to 6).
Variables for the refund vs. CTC analyses	
\$CTC3M	Dollar amount of three months of CTC payments.
\$Refund	Dollar amount of federal tax refund.
Amount	The amount of tax refund (<i>\$Refund</i>) or three months’ CTC payments (<i>\$CTC3M</i>).
CTC(D)	Dummy variable, equal to one for the three-month period after the first CTC monthly payment and zero for the three-month period after refunds.
Overdraft(D)3M	Dummy variable, equal to one for the existence of overdraft/nonsufficient charges to the bank account during the three-month period following tax refunds or the first monthly CTC payments.
Payday_Loan(D)3M	Dummy variable, equal to one if nonzero money flows into the bank account from payday lending or cash advance apps during the three-month period following tax refunds or the first monthly CTC payments.
Gig_Work(D)3M	Dummy variable, equal to one if receiving money from nearly 50 gig platforms such as Uber and Upwork at least once during the three-month period following tax refunds or the first monthly CTC payments.
Cons_Vol\$3M	Consumption volatility, measured as the standard deviation of daily total consumption during the three-month period following tax refunds or the first monthly CTC payments.

Appendix B. Top Merchants by
Consumption Category

The table shows the top 10 merchants by consumption category, excluding cash, based on transaction frequency within category. Cash is primarily ATM/cash withdrawals, without a specific merchant.

Groceries	Other food/dining	Education	Healthcare	Gas
7-Eleven	McDonald’s	Chegg	Walgreens	Shell
Kroger	Starbucks	Teachers Pay Teachers	CVS Pharmacy	Speedway
Wawa	Chick-fil-A	Scholastic Corporation	Rite Aid	Sunoco
Publix	Dunkin’ Donuts	North Gate	Duane Reade	Chevron
Circle K	Taco Bell	Penn State University	Kaiser Permanente	BP Global
Sheetz	Wendy’s	Scribd, Inc.	Express Scripts	Marathon Petroleum
QuikTrip	Burger King	University of North Carolina	Weight Watchers	ExxonMobil
Food Lion	Subway	National Student Clearinghouse	Humana Military	Kroger Fuel Center
H-E-B	Chipotle Mexican Grill	Credentials Solutions	Discount Drug Mart	Pilot Flying J
Safeway	Uber Eats	Course Hero, Inc.	Quest Diagnostics	Costco

Endnotes

¹ For a “fact sheet” summarizing the law, see <https://home.treasury.gov/news/featured-stories/fact-sheet-the-american-rescue-plan-will-deliver-immediate-economic-relief-to-families>. For the full text of the law, see <https://www.congress.gov/bill/117th-congress/house-bill/1319?q=%7B%22search%22%3A%22america+rescue+plan%22%7D&r=1>.

² Baker et al. (2023a) similarly examine the poverty-reduction and labor-supply effects of a change in Canada’s Child Tax Credit policies.

³ Our results of increased grocery and other food/dining spending are consistent with the finding by Parolin et al. (2023) of a drop in food hardship, with a larger drop for low-income recipients. However, they find no rent/mortgage effects, whereas we find significant financial position and liquidity constraint improvements. This difference in results is likely driven by our ability to capture more granular and less sticky measures of liquidity constraints and financial position.

⁴ An important factor to consider with financial aggregator data is how sample individuals are collected (e.g., Jäckle et al. 2021). The data provider liaises with various financial institutions, not directly with individuals. Some institutions employ an opt-out system with customers whereas others employ an opt-in system, depending on their data sharing strategies. Thus, the data do not include all individuals who utilize partner financial institutions. Despite this inherent data limitation, we do not anticipate it to markedly alter our results.

⁵ Envestnet | Yodlee provides the following disclosure regarding their data security and privacy practices: “Envestnet® | Yodlee® follows leading industry practices for data security and privacy. Our data contains no personal identifiers including but not limited to age, gender, ethnicity, address, and healthcare records. For our data analytics offerings, Envestnet | Yodlee imposes measures to protect consumers’ identities, such as prohibiting users from attempting to re-identify any consumer from the data. Importantly, Envestnet | Yodlee does not sell data that identifies consumers.”

⁶ To the extent that consumption and liquidity effects carry over to nonrecipient spouses, this could lead to some understatement of the differences between recipients and matched nonrecipients, as non-recipient spouses can be included in the matched sample.

⁷ We focus on July through September 2021 for practical and conceptual reasons. Extending the sample through December would double the data volume, posing significant computational challenges. Additionally, uncertainty about the CTC’s continuation in late 2021, particularly around the Build Back Better Act, may have influenced recipients’ behavior in various ways (e.g., saving rather than spending). By focusing on the initial three months, we provide a clearer analysis of the CTC’s immediate impact without these additional confounding effects.

⁸ By restricting to multiples or combinations of \$166, \$167, \$250, and \$300, we exclude families with income which falls into the phase-down between \$250/\$300 and \$166/\$167, and the phase-out from \$166/\$167 to \$0. This restriction provides three clean subsamples to examine—lower-income recipients of expanded \$250/\$300 payments, higher-income recipients of \$166/\$167 payments, and nonrecipients who do not have children or have incomes above \$400,000. We match on income tier, which helps to exclude those high-income nonrecipients in our matched control group.

⁹ Payments phase out at lower income levels for parents who file as head of household, are single, or are married filing separately.

¹⁰ The spending amounts in each of the categories we examine include only spending at merchants with a fairly clear focus on the given category. Purchases from multicategory merchants, such as a

department store or big-box store, are classified as “General Merchandise,” and are not included in specific categories. Therefore, our measures may not capture all spending in particular categories.

¹¹ We identify payday borrowing by tracing the money flowing into individuals’ bank accounts from payday lending or cash advance apps (see the list of apps in Appendix A). This probably only accounts for a subset of payday loans as we cannot observe cash transactions if individuals borrow from physical payday loan stores.

¹² Unlike pay cycles for regular jobs where the work and pay could be separated in timing (i.e., employees are likely paid biweekly or monthly at the end of the cycle or with some delay), we can measure gig work participation reasonably well in a one-week window because gig work is typically paid by the week or even more frequently (e.g., Uber drivers are paid weekly but they can request almost real-time pay under Uber’s Instant Pay program).

¹³ We identify gig platforms from an internet search (e.g., <https://investedwallet.com/best-gig-economy-jobs/> and <https://www.thebalancemoney.com/best-sites-to-find-gig-jobs-4150364>).

¹⁴ We do not have data on the number of children an individual has in 2019; thus, it is possible that an individual’s parental status changes between 2019 and 2021. To address the potential impact of this, we replicate the analyses for the subset of enhanced recipients with children aged 6 or older in 2021. We exploit the fact that enhanced recipients receive different payments depending on the age of their children to make this determination. These individuals with children aged 6 or older in 2021 are likely to have had the same children in 2019. Results are robust.

¹⁵ We replicate our main tests using unadjusted consumption, $Ln(1 + RawOutcome_{i,t(2021)})$ and $RawOutcome_{i,t(2021)}$, using both ordinary least squares and Poisson models (Cohn et al. 2022). Results are robust, with generally larger estimates.

¹⁶ Table 3 reports a negative main effect on $Week[+1]$. This implies that spending was lower in $Week[+1]$ than $Week[-1]$, on average, in 2021 than in 2019. Spending levels are often affected by factors such as weather and holiday timing. In this case, the negative main effect is largely driven by September. The key reason is that Labor Day weekend/week had a stronger positive effect on spending in $Week[-1]$ in 2021, when Labor Day was on September 6, compared with 2019, when Labor Day was on September 2. As such, 2021 has a larger drop in consumption from $Week[-1]$ to $Week[+1]$ than 2019 does, for that particular month.

¹⁷ In untabulated analyses, we replicate all tables using entropy balancing. The results are qualitatively similar.

¹⁸ “President Biden and Vice President Harris on Child Tax Credit Payments,” C-SPAN, available at <https://www.c-span.org/video/?513445-1/president-biden-vice-president-harris-child-tax-credit-payments>; Senator Edward Markey on the impact of the CTC, available at <https://www.facebook.com/EdJMarkey/videos/1537473923288337/>.

¹⁹ As with Table 3, we replicate the analyses using raw (unlogged) consumption in each category as the dependent variable, using Poisson models (Cohn et al. 2022). In untabulated results, we find qualitatively similar results.

²⁰ We cannot observe individuals’ AGI. We use \$45,000 as a cutoff as it approximates the lowest AGI at which a taxpayer can receive the full benefit of the minimum CTC without any portion of the credit being refundable.

²¹ As acknowledged in Section 2.1, our data are at the individual level, not household level, and we do not know if the household has a second income earner. Therefore, the income cutoffs we use here do not translate to precise CTC policy thresholds. Nevertheless, the fact that about half of advance-only recipients do not significantly change their spending around CTC payments suggests that

the 2021 expanded CTC might be more generous for higher-income earners than needed, to the extent that the goal of the CTC is to facilitate consumption by families with children.

²² We utilize a different fixed effect structure for these analyses as compared with our main tests. By sample construction, it is not feasible to add state-month fixed effects as tax refunds are received in the first half of 2021 whereas CTC payments are made in the second half. We do not include individual fixed effects as each individual only appears twice in the analyses here.

²³ Because we include tax refunds from January through June, there can be an overlap in the three-month period following the refund and the three-month period of CTC payments. As an additional robustness check, we avoid any overlap by restricting to individuals with a refund received by April 15, 2021. This eliminates 29% of the sample. The results are qualitatively similar for this subsample.

²⁴ We might expect the liquidity and consumption smoothing differences of monthly versus annual payments to be stronger for individuals who are ex ante more liquidity constrained. In additional analyses, we estimate ex ante liquidity constraints using either estimated bank balances, or net interest earned (a proxy for net savings), following Baugh et al. (2021). Using either measure, we find generally stronger incremental effects for monthly payments for individuals who are more liquidity constrained ex ante.

²⁵ The difference in time trend is stronger for individuals who face stronger ex ante liquidity constraints, measured as described in endnote 24. We find a stronger consumption time trend difference between annual and monthly payments for both enhanced and advance-only recipients using both ex ante liquidity constraint measures.

²⁶ Repayments of payday loans account for a small portion of total debt repayment transactions. The results are very similar if we exclude payday loan repayments from debt payments.

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